

SegmentGDA: Training-Free Few-Shot Segmentation via Gaussian Discriminant Analysis on Vision Foundation Features

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Abstract

Few-shot segmentation aims to segment novel object categories given only a handful of annotated examples. Existing approaches either rely on meta-trained specialist models whose learned matching functions generalize poorly across domains or adopt training-free pipelines based on point-wise feature similarity that treat each patch embedding independently and fail to capture the global statistical structure of the target concept. In this paper, we present SegmentGDA, a fully training-free framework that recasts few-shot segmentation as probabilistic inference in the representation space of frozen vision foundation models. SegmentGDA models foreground and background support features as class-conditional Gaussian distributions and applies Gaussian Discriminant Analysis to classify each query patch via Bayesian decision rules, producing a dense posterior probability map that aggregates global class-level statistics across all support samples and suppresses noisy correspondences more effectively than deterministic local matching. The posterior map is further converted into spatially diverse positive and negative point prompts for the Segment Anything Model through an iterative refinement loop, yielding high-resolution segmentation masks without any task-specific training or fine-tuning. Extensive experiments across four standard benchmarks demonstrate the effectiveness of the proposed framework, underscoring its potential for real-world applications. These results highlight distribution-based modeling as a promising approach for adapting vision foundation models to few-shot dense prediction. Code is available at <https://github.com/wavachao/SegmentGDA>.

CCS Concepts

• Computing methodologies → Image segmentation.

*These authors contributed equally to this work.

Keywords

few-shot semantic segmentation, training-free segmentation, vision foundation models, Gaussian discriminant analysis, prompt-based segmentation

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1 Introduction

The rapid progress of deep neural networks has significantly advanced fully supervised semantic segmentation [5, 11, 20]. Despite their impressive performance, these models typically rely on large-scale pixel-level annotations and generalize poorly to unseen categories beyond the training distribution. Few-Shot Segmentation (FSS) [28] has therefore emerged as an important research direction, aiming to segment novel object categories in a query image given only a few annotated support examples (e.g., one or five images with masks). The core challenge of FSS lies in efficiently transferring semantic knowledge from a minimal support set to accurately delineate objects in previously unseen query images, which demands robust generalization well beyond the categories observed during training.

A prevailing paradigm in FSS relies on episodic meta-learning [32, 35, 36], where a specialized segmentation model is trained on base classes to learn transferable feature representations or similarity metrics for novel categories [10, 14, 22, 41]. Representative methods include prototype-based matching [34, 36], dense cross-image correlation modeling [22, 25, 30], and feature-level prompting [42]. Although these meta-trained specialist models have established strong benchmarks on in-domain evaluation, they often suffer from two key limitations: (1) limited cross-domain generalization, as the learned matching functions are inherently biased toward the base-class distribution encountered during training [19]; and (2) heavy reliance on large-scale densely annotated data for meta-training, which restricts scalability and practical deployment to new domains or application scenarios.



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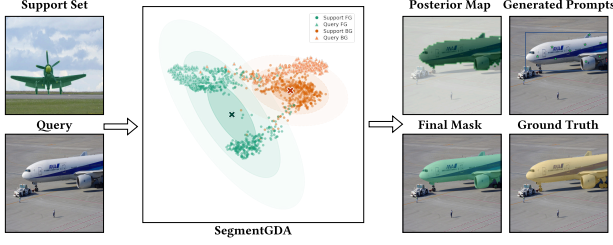


Figure 1: Motivation of SegmentGDA. The left panel shows the support images alongside a query image. In the middle panel, foundation features are projected into a 2D PCA space, where dark teal and light green points denote support and query foreground features, while dark orange and light orange points denote support and query background features. The stars and ellipses indicate the estimated means and covariance contours of the class-conditional Gaussian distributions. The right panel shows the derived dense posterior map, the automatically generated positive and negative point prompts, and the final segmentation mask.

Recently, vision foundation models (VFMs), including vision-language models such as CLIP [26], promptable segmentation models such as SAM [12, 27], and self-supervised feature encoders from the DINO family, such as DINOv2 [24] and DINOv3 [31], have demonstrated remarkable generalization capabilities across diverse visual tasks without task-specific retraining. These advances have inspired a new line of research that leverages foundation models as generalist segmentation frameworks for FSS [37, 38]. Instead of training task-specific architectures, several recent training-free approaches directly exploit pretrained visual features to establish semantic correspondences between support and query images, typically through point-wise similarity measures such as cosine similarity or bipartite matching [18, 29, 43].

These matching-based strategies exhibit promising cross-domain robustness; however, they share a fundamental limitation: each pixel or patch embedding is treated as an independent sample, and classification relies solely on local pairwise similarities. As a result, these methods fail to capture the *global statistical structure* of the target concept, which limits their ability to model intra-class variability and suppress background noise. This limitation is particularly pronounced when the target object exhibits large appearance variations or structural fragmentations across images; in such cases, relying on isolated, unconstrained point-to-point matches often leads to severe false positives in cluttered backgrounds and incomplete coverage of the target object. To the best of our knowledge, prior training-free few-shot segmentation methods have not explicitly formulated dense support-to-query transfer via Gaussian Discriminant Analysis in the frozen feature space of a vision foundation model.

To overcome these limitations, we propose a paradigm shift from deterministic point-wise matching to probabilistic distribution modeling, as conceptualized in Figure 1. Although the query remains semantically ambiguous from raw appearance in the few-shot setting, jointly considering the support and query representations suggests

that semantic transfer is better characterized by class-conditional Gaussian statistics than by isolated point-wise correspondences, naturally motivating Gaussian Discriminant Analysis. Specifically, when projected into a Principal Component Analysis (PCA) space (Figure 1, middle), the dark teal points denote foreground features from the support set, while the light green points denote foreground features from the query image; correspondingly, the dark orange and light orange points represent support and query background features. The stars indicate the estimated class means, and the ellipses visualize the covariance structure of the corresponding Gaussian distributions. By viewing foreground and background features as class-conditional distributions in the foundation feature space, the model can derive a dense posterior map over the query image and convert it into reliable prompts for mask generation.

In this work, we revisit the representation space of vision foundation models from a statistical perspective. Empirically, we observe that high-dimensional features extracted by models such as DINOv2 for a given semantic category exhibit approximately multivariate Gaussian distributions in feature space. This observation is consistent with prior findings in out-of-distribution detection [15], where class-conditional Gaussian modeling of deep features has been shown to yield effective Mahalanobis-based classifiers, suggesting that the Gaussian assumption is a natural and well-supported inductive bias for deep visual representations. Modeling the target concept as a probabilistic distribution, rather than relying on deterministic point-wise matching, enables a more robust representation that naturally aggregates information across all support samples and mitigates the influence of noisy or atypical features.

Based on this observation, we propose **SegmentGDA**, a fully training-free few-shot segmentation framework built upon Gaussian Discriminant Analysis (GDA) [2, 23]. Given a small set of support images, SegmentGDA first extracts dense visual features using a frozen foundation encoder and estimates class-conditional Gaussian distributions for both foreground and background regions. For a query image, instead of computing nearest-neighbor similarities, our method applies Bayesian decision rules by evaluating the Mahalanobis distance [21] with respect to the estimated distributions. This probabilistic inference produces a dense posterior probability map indicating the likelihood that each query patch belongs to the target class. We further transform these posterior probabilities into reliable positive and negative point prompts, which are subsequently provided to SAM to obtain the final high-resolution segmentation mask. The entire pipeline operates solely through inference with frozen models, requiring no task-specific training, fine-tuning, or learnable adaptation at any stage.

We evaluate SegmentGDA on four widely used few-shot segmentation benchmarks: COCO-20ⁱ [17], FSS-1000 [16], LVIS-92ⁱ [9], and PASCAL-5ⁱ [7]. Our experiments show that SegmentGDA provides a robust and effective framework for few-shot segmentation, achieving highly competitive performance on standard benchmarks. These results highlight the effectiveness and robustness of distribution-based modeling, providing a mathematically principled and highly interpretable alternative for adapting foundation features toward few-shot segmentation.

Our main contributions are summarized as follows:

- We propose **SegmentGDA**, a training-free few-shot segmentation framework that performs dense support-to-query transfer via Gaussian Discriminant Analysis in the frozen feature space of a vision foundation model. To the best of our knowledge, SegmentGDA is the first framework to introduce this formulation, moving beyond local point-wise matching.
- We introduce a probabilistic prompt generation strategy that bridges statistical distribution modeling and foundation segmentation models via Bayesian inference, eliminating the need for task-specific training or fine-tuning.
- We design an iterative mask refinement mechanism driven by the GDA posterior probability, which extracts spatially distinct positive and negative prompts to systematically correct false predictions, enabling highly accurate object boundary delineation even in cluttered scenes.
- Extensive experiments on COCO-20ⁱ, FSS-1000, LVIS-92ⁱ, and PASCAL-5ⁱ validate SegmentGDA as an effective and robust paradigm for training-free few-shot segmentation, demonstrating strong performance across multiple benchmarks.

2 Related Work

Meta-Learning Based Few-Shot Segmentation. Few-shot semantic segmentation (FSS) has been extensively studied under the episodic meta-learning paradigm [28, 35]. Early prototype-based approaches [6, 34, 36] represent each class by a single global prototype and classify query features via cosine distance, providing a simple yet effective baseline for the few-shot setting. Subsequent work has moved beyond single prototypes toward richer support-query interaction. HSNet [22] builds dense 4D hypercorrelation tensors to capture multi-level feature correspondences; VAT [10] introduces a cost aggregation transformer that reasons over the full 4D cost volume; DCAMA [30] employs dense cross-attention to perform fine-grained support-query matching; HDMNet [25] decomposes matching into hierarchical patch- and pixel-level stages for multi-scale correlation distillation; and FPTrans [42] reformulates FSS as visual token-level prompting within a transformer encoder. Although these specialist models achieve strong in-domain performance, their success hinges on substantial supervised meta-training on base classes and architectures tailored to the episodic setting. Consequently, their generalization to unseen domains or categories is often constrained by the base-class distribution, and adapting them to new data regimes remains costly [19]. Our work departs from this paradigm entirely: instead of learning task-specific matching functions, SegmentGDA exploits the intrinsic statistical structure of pretrained foundation features without any training.

Vision Foundation Models for Segmentation. Vision foundation models have opened a fundamentally different path for few-shot segmentation by providing strong, transferable visual representations. Generalist frameworks such as Painter [37] and SegGPT [38] cast segmentation as an in-context visual prediction problem, demonstrating that large pretrained models can perform dense prediction with minimal task-specific adaptation. In parallel, self-supervised encoders such as DINO [3] and its successor DINOv2 [24] have shown that patch-level features learned without labels capture rich semantic structure amenable to downstream dense prediction tasks.

On the segmentation side, promptable models such as SAM [12] and SAM 2 [27] accept flexible visual prompts (points, boxes, or masks) and produce high-quality segmentation masks, making them natural decoders for training-free pipelines. While these foundation models provide powerful building blocks, generalist inference alone is often insufficient for precise few-shot mask transfer: the model lacks a mechanism to identify which concept the user intends to segment when only a handful of annotated support examples are given.

Foundation-Model-Based Few-Shot Segmentation. A growing line of work, therefore, leverages foundation models with explicit support-query interaction, prompt engineering, or lightweight adaptation for few-shot segmentation. Matcher [18] establishes dense feature correspondences via bipartite matching and converts the matched regions into SAM prompts, achieving competitive results without training. Among training-free methods, PerSAM [43] explores personalized one-shot prompting by localizing the target through a single support feature and deriving point and box prompts for SAM; VRP-SAM [33] feeds visual reference prompts directly into SAM for reference-guided segmentation; UNICL-SAM [29] introduces uncertainty-driven in-context learning with part prototype discovery to handle fine-grained segmentation under ambiguity; and GF-SAM [40] models relations among prompt points with a graph-based design to reduce prompt ambiguity in training-free FSS. In contrast to the training-free methods above, FS-SAM2 [8] repurposes SAM 2’s video-memory mechanism for FSS and meta-trains LoRA modules in its image encoder and memory components. The same study also reports VRP-SAM2, an implementation of VRP-SAM [33] that replaces SAM with SAM 2 as the segmentation backbone.

Despite their practical effectiveness, most of these training-free methods still rely on deterministic point-wise similarity, prototype matching, or heuristic prompt construction: they compare individual patch features between support and query images and make local, independent decisions. This design ignores the global distributional structure of the target category, limiting its capacity to model intra-class variance and to distinguish the target from semantically similar background regions. While distribution-based modeling has shown promise for few-shot image classification [1], its application to training-free few-shot dense prediction over frozen foundation features remains largely unexplored. To the best of our knowledge, none of these training-free FSS methods explicitly formulate dense support-to-query transfer via Gaussian Discriminant Analysis in the frozen feature space of a vision foundation model. In contrast, SegmentGDA models support features as class-conditional Gaussian distributions and perform Bayesian inference in the representation space of a frozen vision foundation model. Rather than making correspondence decisions from isolated local matches, our method captures the global statistics of foreground and background features through Gaussian Discriminant Analysis, yielding more robust posterior estimation and more reliable prompts for the final SAM decoding stage.

3 Method

In this section, we present **SegmentGDA**, a fully training-free framework for few-shot semantic segmentation. The key idea is

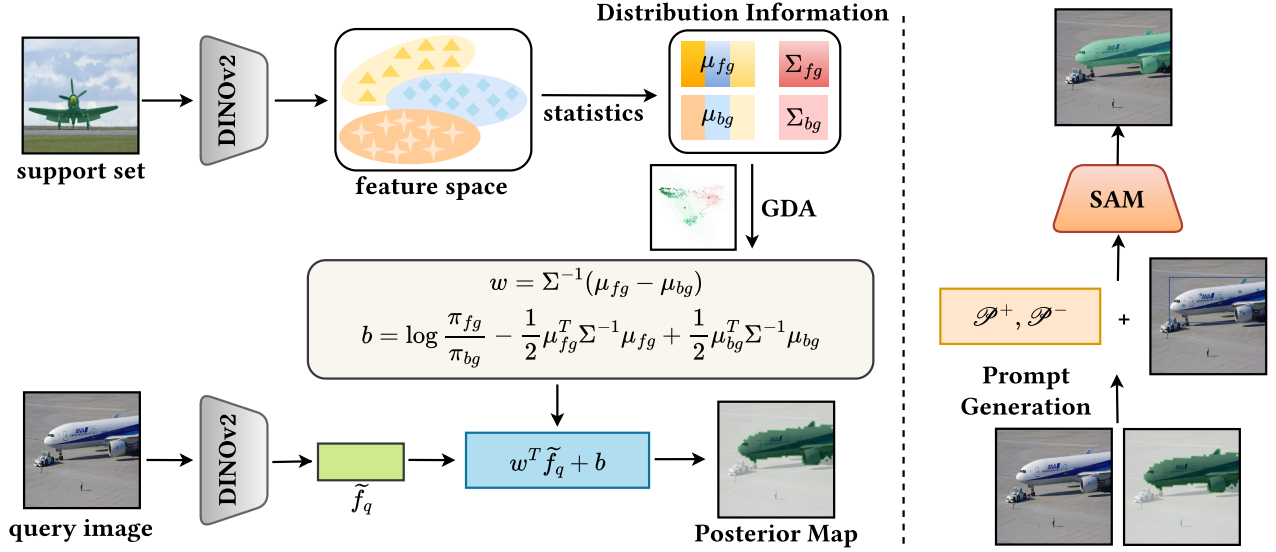


Figure 2: Overview of SegmentGDA. A frozen DINOv2 encoder extracts dense patch features from the support set and the query image; all features are then power-transformed as in Eq. (1). The support foreground and background features are used to estimate the shared-covariance Gaussian parameters (Eq. (3)), from which the linear discriminant weight vector w and scalar bias b are derived (Eq. (4)). For each query patch, the foreground posterior probability is computed via $\sigma(w^T \tilde{f}_q + b)$ (Eq. (5)), producing a dense posterior map. This map guides the generation of probabilistic point prompts ($\mathcal{P}^+, \mathcal{P}^-$) to drive SAM for mask decoding and iterative refinement.

to model the dense visual features of foreground and background as class-conditional Gaussian distributions and perform Bayesian classification at the patch level to produce a dense posterior map. This map then serves as the basis for generating point prompts that guide a promptable segmentation model to produce the final high-resolution mask.

As illustrated in Figure 2, the pipeline consists of three stages. First, a frozen vision foundation model extracts dense patch-level features from both support and query images (§3.1). Second, we fit a Gaussian Discriminant Analysis (GDA) model on the support features and apply it to the query features, yielding a per-patch foreground posterior map (§3.2). Third, we perform posterior-guided segmentation by converting this map into probabilistic point prompts that drive the Segment Anything Model (SAM) for high-resolution mask decoding (§3.3).

3.1 Dense Feature Extraction

Given a support set $\mathcal{S} = \{(I_s^{(k)}, M_s^{(k)})\}_{k=1}^K$ consisting of K image-mask pairs, where $I_s^{(k)} \in \mathbb{R}^{H \times W \times 3}$ is an RGB support image and $M_s^{(k)} \in \{0, 1\}^{H \times W}$ is its corresponding binary mask, and a query image $I_q \in \mathbb{R}^{H \times W \times 3}$, we encode every image using a frozen DINOv2 [24] encoder as a dense feature extractor. The encoder outputs a grid of patch-level feature vectors for each image; we discard the class (CLS) token [24] to retain only the spatial patch features.

To better satisfy the Gaussian assumption in the subsequent modeling stage, we apply a power transform (Tukey’s Ladder of Powers [4]) followed by ℓ_2 normalization to each raw patch feature

$\mathbf{f} \in \mathbb{R}^D$:

$$\tilde{\mathbf{f}} = \frac{\text{sgn}(\mathbf{f}) \odot \sqrt{|\mathbf{f}|}}{\|\text{sgn}(\mathbf{f}) \odot \sqrt{|\mathbf{f}|}\|_2}, \quad (1)$$

where $\tilde{\mathbf{f}}$ denotes the transformed feature, $\odot$ denotes the Hadamard product, and $\text{sgn}(\cdot)$ and $|\cdot|$ operate element-wise. This square-root transform suppresses heavy-tailed outliers in the foundation model’s representation space, yielding more compact, approximately Gaussian feature distributions that are better suited to parametric modeling.

3.2 Gaussian Discriminant Analysis

Rather than matching query patches to individual support features, SegmentGDA aggregates all support patches into global class-level statistics and frames segmentation as a binary Bayesian classification problem. We assume that the transformed features of the foreground and background each follow a multivariate Gaussian distribution with a shared covariance matrix, corresponding to the assumption of Linear Discriminant Analysis (LDA) [2, 23].

Parameter Estimation. Using the support masks, we partition the transformed support features into a foreground set $\mathcal{F}_{\text{fg}}$ and a background set $\mathcal{F}_{\text{bg}}$, with their cardinalities denoted as $N_{\text{fg}} = |\mathcal{F}_{\text{fg}}|$ and $N_{\text{bg}} = |\mathcal{F}_{\text{bg}}|$, respectively. We compute their class means $\mu_{\text{fg}}, \mu_{\text{bg}} \in \mathbb{R}^D$ by sample averaging. Let $S_c \in \mathbb{R}^{D \times D}$ denote the unnormalized within-class scatter matrix for class $c \in \{\text{fg}, \text{bg}\}$. It is

computed as:

$$S_c = \sum_{\tilde{\mathbf{f}} \in \mathcal{F}_c} (\tilde{\mathbf{f}} - \boldsymbol{\mu}_c)(\tilde{\mathbf{f}} - \boldsymbol{\mu}_c)^\top, \quad c \in \{\text{fg}, \text{bg}\}, \quad (2)$$

and the pooled covariance matrix with ridge regularization [13] for numerical stability is:

$$\Sigma = \frac{S_{\text{fg}} + S_{\text{bg}}}{N_{\text{fg}} + N_{\text{bg}} - 2} + \lambda \mathbf{I}, \quad (3)$$

where $\mathbf{I} \in \mathbb{R}^{D \times D}$ is the identity matrix, and λ is a ridge regularization coefficient to ensure positive definiteness in high-dimensional feature spaces. The class prior probabilities π_{fg} and π_{bg} are estimated from the proportion of foreground and background patches in the support set.

Linear Discriminant Function. Under the shared-covariance Gaussian assumption, the log-posterior ratio between the two classes reduces to a linear function of the feature vector [2, 23]. As shown in Figure 2, this yields a weight vector $\mathbf{w} \in \mathbb{R}^D$ and a scalar bias $b \in \mathbb{R}$:

$$\mathbf{w} = \Sigma^{-1}(\boldsymbol{\mu}_{\text{fg}} - \boldsymbol{\mu}_{\text{bg}}), b = \log \frac{\pi_{\text{fg}}}{\pi_{\text{bg}}} - \frac{1}{2} \boldsymbol{\mu}_{\text{fg}}^\top \Sigma^{-1} \boldsymbol{\mu}_{\text{fg}} + \frac{1}{2} \boldsymbol{\mu}_{\text{bg}}^\top \Sigma^{-1} \boldsymbol{\mu}_{\text{bg}}, \quad (4)$$

A similar linear discriminant formulation has recently been shown to provide a strong baseline for training-free adaptation of vision-language models [39], further validating the effectiveness of Gaussian-based linear classifiers in pretrained feature spaces.

For each patch in the query image, we apply the same power transformation (Eq. (1)) to its raw feature $\mathbf{f}_q$, yielding the transformed query feature $\tilde{\mathbf{f}}_q \in \mathbb{R}^D$. The foreground posterior probability is then:

$$P(\text{fg} | \tilde{\mathbf{f}}_q) = \sigma(\mathbf{w}^\top \tilde{\mathbf{f}}_q + b). \quad (5)$$

where $\sigma(z) = 1/(1 + \exp(-z))$ is the sigmoid function. Evaluating this for every query patch produces a dense posterior map that indicates the per-location likelihood of belonging to the target class.

3.3 Posterior-Guided Segmentation

The posterior map provides a reliable but spatially coarse localization of the target object, limited by the patch resolution of the encoder. To recover fine-grained object boundaries, we use it to generate point prompts for a promptable segmentation model such as SAM [12].

Prompt Sampling. To construct a robust set of guidance signals for SAM, we identify high-confidence foreground and background regions by thresholding the posterior map. Within each region, instead of relying on a single spatial center, we apply k -means++ initialization to extract a spatially diverse set of positive prompts $\mathcal{P}^+$ and negative prompts $\mathcal{P}^-$. This spatial diversity ensures that the prompts cover various object parts while actively suppressing surrounding clutter. To further improve robustness against prompt ambiguity, we randomly draw multiple variable-sized subsets from $\mathcal{P}^+ \cup \mathcal{P}^-$ and decode each subset independently through SAM, producing a pool of candidate masks from which the best one is selected.

Table 1: Results (mIoU %) on COCO-20ⁱ, FSS-1000, and LVIS-92ⁱ. Best and second-best training-free results are bolded and underlined, respectively. Methods shown in gray require additional training or fine-tuning.

Method	COCO-20 ⁱ		FSS-1000		LVIS-92 ⁱ	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
<i>Specialist models</i>						
HSNet [22]	41.2	49.5	86.5	88.5	17.4	22.9
VAT [10]	41.3	47.9	90.3	90.8	18.5	22.7
FPTans [42]	47.0	58.9	-	-	-	-
<i>Generalist models</i>						
Painter [37]	33.1	32.6	61.7	62.3	10.5	10.9
SegGPT [38]	56.1	67.9	85.6	89.3	18.6	25.4
FS-SAM2 [8]	55.3	58.4	90.6	91.7	-	-
PerSAM [43]	23.5	-	75.6	-	12.3	-
Matcher [18]	52.7	60.7	<u>87.0</u>	<u>89.6</u>	33.0	40.0
UNICL-SAM [29]	77.8	78.7	84.0	86.3	<u>34.1</u>	37.4
GF-SAM [40]	<u>58.7</u>	<u>66.8</u>	88.0	88.9	35.2	44.2
SegmentGDA	42.1	62.5	82.9	89.8	26.2	<u>41.8</u>

Iterative Refinement. To systematically correct false predictions, we introduce an iterative feedback loop between the GDA posterior map and the SAM decoder. By comparing the initial SAM prediction against the dense posterior map, we identify confident false negatives (regions with high posterior probability that are absent from the mask) and false positives (regions with low posterior probability that are erroneously included in the mask). These diagnostic error regions are converted into supplementary point prompts and fed back into SAM for a subsequent decoding pass. This iterative refinement process refines ambiguous boundaries and recovers missing components, enabling highly accurate object delineation even in severely cluttered scenes without requiring any meta-training.

4 Experiments

4.1 Experimental Setup

Task Formulation. Few-shot semantic segmentation (FSS) aims to segment novel object classes given only a handful of annotated examples. Following the standard episodic evaluation protocol [14, 28, 35, 36], each episode samples a support set $\mathcal{S} = \{(I_s^{(k)}, M_s^{(k)})\}_{k=1}^K$ of K image-mask pairs and a query image I_q , all sharing the same target class. The model must produce a binary segmentation mask for I_q using only the information in $\mathcal{S}$, without access to any training data of the target class. We consider both the 1-shot ($K = 1$) and 5-shot ($K = 5$) settings and report mean Intersection over Union (mIoU) averaged over all classes and folds.

Datasets. We evaluate on four standard FSS benchmarks. **PASCAL-5ⁱ** [7] and **COCO-20ⁱ** [17] are widely used general-purpose benchmarks constructed from PASCAL VOC 2012 and MS COCO, respectively, each cross-validated over 4 folds. **FSS-1000** [16] covers 1000 diverse object classes, testing generalization to fine-grained and unconstrained concepts. **LVIS-92ⁱ** [9] is built upon the LVIS

Table 2: FSS performance on the PASCAL-5ⁱ dataset (mIoU metric). Best results in bold, second-best in underlined.

Method	PASCAL-5 ⁱ 1-shot					PASCAL-5 ⁱ 5-shot				
	fold-0	fold-1	fold-2	fold-3	mean	fold-0	fold-1	fold-2	fold-3	mean
PFENet [34]	61.7	69.5	55.4	56.3	60.8	63.1	70.7	55.8	57.9	61.9
HSNet [22]	64.3	70.7	60.3	60.5	64.0	70.3	73.2	67.4	67.1	69.5
DCAMA [30]	72.2	73.8	64.3	<u>67.1</u>	69.4	75.7	77.1	72.0	74.8	74.9
FPTTrans [42]	72.3	70.6	68.3	64.1	68.8	76.7	79.0	<u>81.0</u>	75.1	78.0
Matcher [18]	67.7	70.7	66.9	67.0	68.1	71.4	77.5	74.1	72.8	74.0
HDMNet [25]	71.0	75.4	68.9	62.1	69.4	71.3	76.2	71.3	68.5	71.8
VRP-SAM [33]	<u>73.9</u>	<u>78.3</u>	<u>70.6</u>	65.0	71.9	-	-	-	-	72.9
VRP-SAM2 [†] [8]	74.0	77.4	69.3	63.5	71.0	75.6	74.6	69.7	61.8	70.4
FS-SAM2 [8]	74.0	79.6	73.5	66.7	73.4	75.7	79.6	75.3	68.8	74.8
GF-SAM [40]	71.1	75.7	69.2	73.3	<u>72.1</u>	81.5	86.3	79.7	82.9	82.6
SegmentGDA	55.3	62.9	60.6	59.0	59.5	<u>79.9</u>	<u>83.0</u>	81.7	<u>84.2</u>	<u>82.2</u>

[†] VRP-SAM reimplemented with SAM 2 as the backbone, as reported in [8].

dataset with massive vocabulary and natural long-tail class distributions, serving as a challenging testbed for open-vocabulary generalization.

Implementation Details. We use DINOv2 ViT-L/14 [24] as the frozen feature encoder and SAM ViT-H [12] as the promptable segmentor. No training, fine-tuning, or learnable adaptation is involved at any stage. To ensure fair comparability, our evaluation protocols—including the choice of foundation model backbones, input image sizes (518×518 for DINOv2 and 1024×1024 for SAM), dataset splits, metrics (mIoU), and extensive episodic sampling—strictly follow Matcher [18]. While our Bayesian framework exhibits strong statistical robustness across domains, we specify all auxiliary algorithmic constants for full reproducibility. During GDA modeling, we employ a single Gaussian component and a covariance regularization scale of 10^{-4} . For iterative prompt generation, the point extraction threshold is set to 0.68 with a maximum of 10 point prompts. These parameters remain strictly frozen across all evaluated benchmarks without any dataset-specific tuning. All experiments are conducted on a single NVIDIA RTX 2080 Ti GPU.

Baselines. We compare against two families of methods. *Specialist models* are meta-learning architectures trained end-to-end on base classes to learn transferable segmentation priors: PFENet [34] enriches prototypes with multi-scale prior masks; HSNet [22] builds dense 4D correlation tensors between support-query feature pairs; DCAMA [30] introduces cross-attention for dense matching; FPTTrans [42] formulates FSS as visual token-level prompting; and HDMNet [25] decomposes matching into patch- and pixel-level stages. *Generalist models* build on large-scale pretrained or foundation models and transfer them to FSS through in-context prediction, prompt engineering, dense correspondence, or lightweight adaptation. Painter [37] and SegGPT [38] rely on pretrained in-context segmentation models; PerSAM [43], Matcher [18], VRP-SAM [33], UNICL-SAM [29], and GF-SAM [40] perform training-free transfer from frozen visual features or prompts; and FS-SAM2 [8] adapts SAM 2 to FSS through LoRA-based meta-training of its image and memory components. We additionally report VRP-SAM2, the SAM 2-backbone implementation of VRP-SAM reported in the same study.

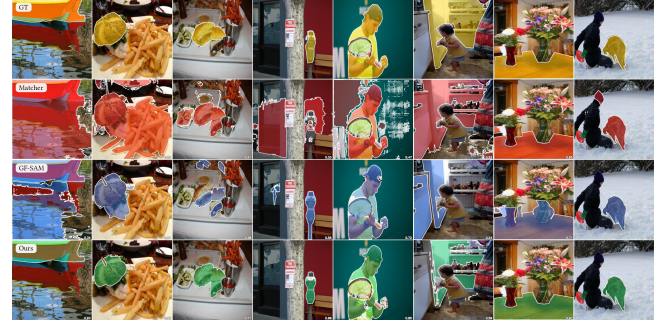


Figure 3: Qualitative comparison on selected 5-shot COCO-20¹ episodes. Each column corresponds to one episode, and the rows show the ground-truth mask, Matcher, GF-SAM, and SegmentGDA, respectively. IoU scores are overlaid on the predicted masks.



Figure 4: Qualitative comparison on selected 5-shot FSS-1000 episodes. Each column corresponds to one episode, and the rows show the ground-truth mask, Matcher, GF-SAM, and SegmentGDA, respectively. IoU scores are overlaid on the predicted masks.

4.2 Main Results

Results on COCO-20ⁱ, FSS-1000, and LVIS-92ⁱ. Table 1 summarizes the comparison on three large-scale benchmarks. In the 5-shot setting, SegmentGDA achieves strong performance among generalist models on both FSS-1000 (89.8%) and LVIS-92ⁱ (41.8%), while remaining competitive on COCO-20ⁱ (62.5%). On LVIS-92ⁱ, which features a challenging long-tail class distribution with 92 categories, SegmentGDA outperforms several recent generalist baselines, including Matcher by 1.8% and UNICL-SAM by 4.4% in the 5-shot setting, supporting the advantage of distribution-based modeling under high intra-class variation. On FSS-1000, SegmentGDA reaches 89.8% in the 5-shot setting, outperforming Matcher (89.6%), UNICL-SAM (86.3%), and GF-SAM (88.9%). A plausible explanation is that many FSS-1000 episodes exhibit relatively coherent target semantics and clearer foreground-background separation, allowing global class-conditional distribution modeling to aggregate support evidence more effectively than purely local correspondence matching. This advantage becomes most evident in the multi-shot regime, where multiple support examples yield more stable estimates of the foreground and background statistics.

We note that SegmentGDA’s 1-shot performance is comparatively modest (e.g., 42.1% on COCO-20ⁱ). This is an inherent characteristic of our statistical framework: with only a single support image, the number of available foreground patches is limited, leading to a class-conditional mean that may not be representative of the true class centroid and a covariance estimate that is dominated by the regularization term. In contrast, methods such as UNICL-SAM employ part-level prototype mining and contrastive learning that can extract discriminative cues from a single exemplar more effectively. As the number of support samples increases, our distributional estimates become increasingly accurate, which is validated by the substantial 1-shot $\rightarrow$ 5-shot improvement across all benchmarks (+20.4% on COCO-20ⁱ, +6.9% on FSS-1000, +15.6% on LVIS-92ⁱ).

Results on PASCAL-5ⁱ. Table 2 reports the fold-wise comparison on PASCAL-5ⁱ. In the 5-shot setting, SegmentGDA achieves an average mIoU of 82.2%, surpassing all specialist baselines and remaining highly competitive with recent generalist models on PASCAL-5ⁱ. This result exceeds the strong meta-learning method FPTrans by 4.2%, improves over the generalist baseline Matcher (74.0%) by a substantial margin, and remains within 0.4% of GF-SAM (82.6%). In the 1-shot setting, SegmentGDA (59.5%) trails FS-SAM2 (73.4%) and other methods that benefit from task-specific prompt engineering or SAM 2’s memory-bank mechanism. However, the dramatic improvement from 59.5% (1-shot) to 82.2% (5-shot) confirms the strong scalability of our distribution-based approach: the 22.7% gain is the largest 1-shot $\rightarrow$ 5-shot improvement among all compared methods, indicating that SegmentGDA benefits particularly strongly from additional support observations.

4.3 Qualitative Results

Figure 3 and Figure 4 compare SegmentGDA with Matcher [18] and GF-SAM [40], two representative generalist baselines based on dense correspondence and graph-guided prompt reasoning, on selected challenging 5-shot episodes from COCO-20ⁱ and FSS-1000, respectively. In each figure, columns correspond to episodes,

Table 3: Ablation results (mIoU %) on COCO-20ⁱ 5-shot.

Ablation	Variant	mIoU
<i>(a) Effectiveness of Classification Strategy</i>		
	Point-wise matching	53.6
	Shared diagonal covariance	60.8
	Mean-only Euclidean	59.9
	Shared full covariance (default)	62.5
<i>(b) Impact of Distributional Complexity</i>		
	3-component GMM ($K = 3$)	61.0
	2-component GMM ($K = 2$)	61.7
	Single Gaussian ($K = 1$, default)	62.5
<i>(c) Effectiveness of Feature Dimensionality</i>		
	PCA-32	55.6
	PCA-64	57.7
	PCA-128	59.0
	PCA-256	59.4
	No PCA, 1024-d (default)	62.5

while the rows show the ground-truth mask, Matcher prediction, GF-SAM prediction, and SegmentGDA prediction; IoU scores are directly overlaid on the predicted masks for visual comparison. Across diverse categories and scene contexts, SegmentGDA produces masks with stronger object-level consistency and cleaner foreground-background separation than both baselines. In particular, our method is less prone to false positives in cluttered background regions and better preserves fine structures such as thin appendages, fragmented parts, and occlusion boundaries, owing to the global distributional inference that suppresses noisy local correspondences and yields more reliable prompts for the final SAM decoding stage.

Figure 5 further visualizes the full inference pipeline of SegmentGDA on five representative COCO-20ⁱ episodes. For each episode, the four columns show, from left to right: the original query image, the GDA posterior map overlaid on the query with generated point prompts (green stars for positive, red crosses for negative), the resulting segmentation mask, and the ground truth. The posterior maps demonstrate that the GDA classifier assigns high probability to the target foreground regions while suppressing background, and the derived point prompts are spatially well-distributed across the target object, enabling SAM to produce accurate segmentation masks. These visual results corroborate the quantitative findings: the improvement is most evident in multi-shot episodes, where richer support observations lead to more accurate class-conditional statistics and hence more reliable posterior-based prompt generation for SAM.

4.4 Ablation Studies

We conduct ablation experiments on COCO-20ⁱ in the 5-shot setting to analyze the contribution of each design choice. All results are averaged over 4 folds.

Effectiveness of Classification Strategy. Table 3(a) compares four classification strategies. The point-wise matching baseline (i.e., dense cosine similarity nearest-neighbor matching), which follows

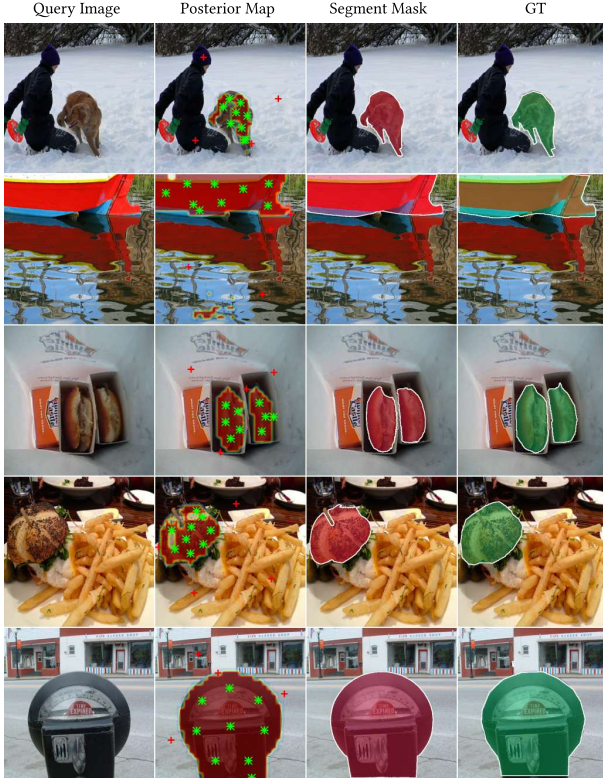


Figure 5: Visualization of the SegmentGDA inference pipeline on five COCO-20ⁱ episodes. Each row shows one episode with four columns: the query image, the GDA posterior map overlaid on the query with point prompts (green stars = positive, red crosses = negative), the predicted segmentation mask (red overlay), and the ground-truth mask (green overlay).

the local correspondence paradigm of existing training-free methods, achieves 53.6%. Switching to any form of global statistics-based classification yields a substantial improvement of at least +6%, regardless of whether a shared diagonal covariance, shared full covariance, or simple Euclidean distance to the class mean is used. This result confirms that the primary advantage of SegmentGDA lies in aggregating support features into global class-level statistics rather than relying on per-patch local similarities. The shared full covariance variant achieves the best result (62.5%), validating the benefit of modeling the complete second-order structure of the feature distribution. The comparable performance of the three global variants (59.9–62.5%) suggests that the dominant factor is the shift from local to global modeling, while the specific covariance parameterization provides an additional but moderate gain.

Impact of Distributional Complexity. Table 3(b) examines the impact of modeling the target feature distribution with varying degrees of complexity, comparing Multi-Component Gaussian Mixture Models (GMM) with $K \in \{2, 3\}$ components against our default

single-component Gaussian ($K = 1$). Surprisingly, increasing the number of components from $K = 1$ to $K = 3$ steadily degrades performance from 62.5% to 61.0%. We hypothesize that in the data-scarce few-shot regime, partitioning the already limited support features into multiple sub-clusters exacerbates sample sparsity. This leads to unstable or degenerate intra-cluster covariance estimation, which ultimately distorts the Bayesian decision boundary. Conversely, aggregating all features into a single global Gaussian maximally exploits the available support observations. This finding solidly validates our core theoretical design: an elegant single-component representation combined with a robust pooled covariance estimator provides the most effective statistical prior for dense feature modeling.

Effectiveness of Feature Dimensionality. Table 3(c) examines the effect of Principal Component Analysis (PCA) based dimensionality reduction. Performance degrades monotonically as the feature dimension decreases, dropping from 62.5% at the full 1024 dimensions to 55.6% at 32 dimensions. This suggests that the DINOv2 features are information-dense and exhibit low redundancy; even moderate linear compression discards fine-grained visual details critical for distinguishing novel classes. SegmentGDA therefore operates in the full feature space and relies on robust ridge regularization (Eq. (3)) to handle the high dimensionality, striking a balance between discriminative power and numerical stability.

5 Conclusion

In this paper, we introduced **SegmentGDA**, a training-free few-shot segmentation framework that performs dense support-to-query transfer via Gaussian Discriminant Analysis in the frozen feature space of a vision foundation model. To the best of our knowledge, SegmentGDA is the first framework to introduce this formulation, moving beyond local point-wise matching. By treating dense patch features extracted from a frozen DINOv2 encoder as samples from class-conditional Gaussian distributions, SegmentGDA aggregates global statistics from a small set of support examples and applies Gaussian Discriminant Analysis to classify each query patch via Bayesian posterior estimation. The resulting dense posterior map is then converted into spatially diverse point prompts for SAM-based mask decoding, with an iterative refinement loop that corrects false positives and false negatives through diagnostic feedback. Extensive experiments across diverse benchmarks highlight the practical utility of our approach and underscore the value of distribution-based modeling as a principled and interpretable paradigm for reliable few-shot dense prediction.

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